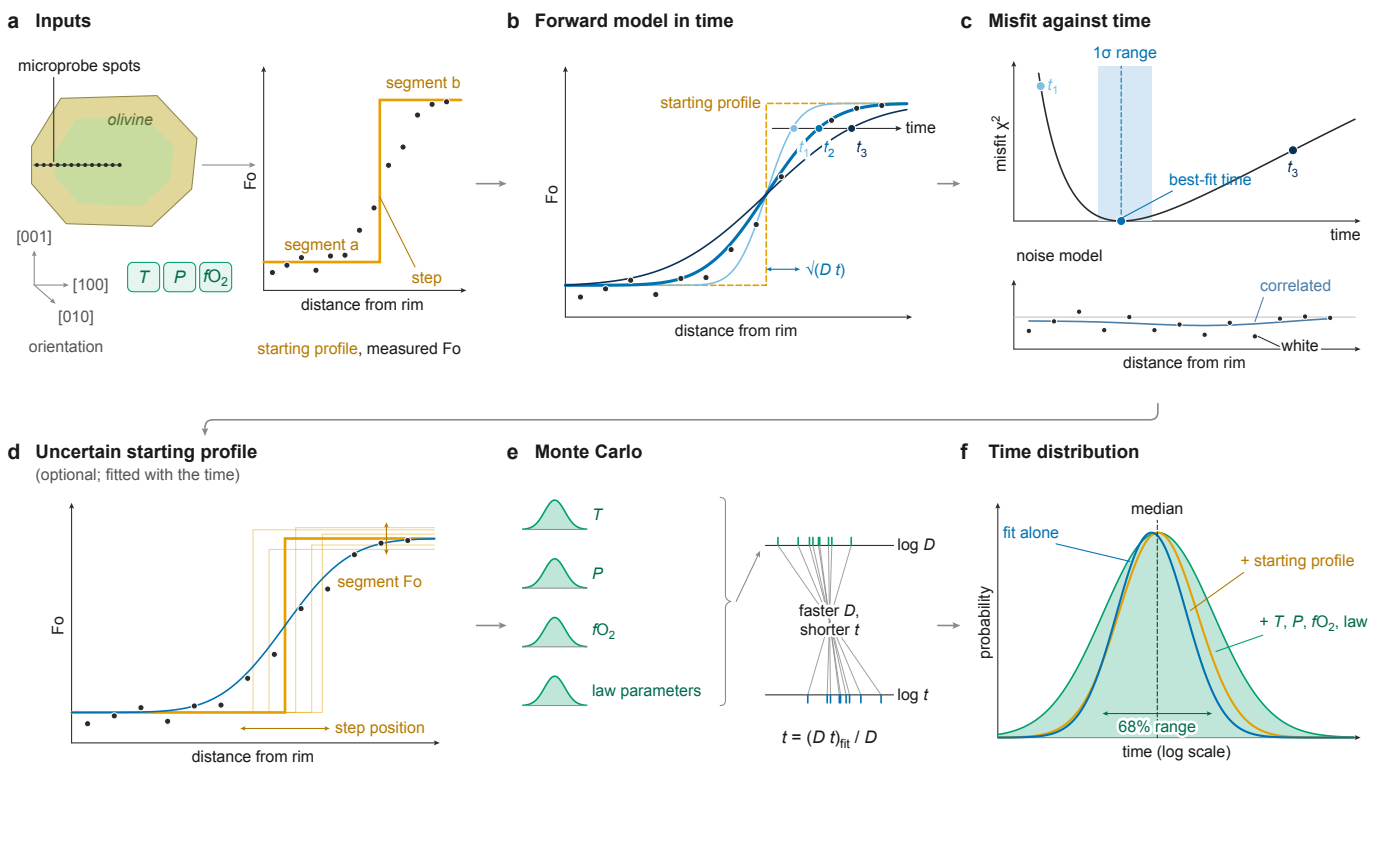

a Inputs
microprobe spots
olivine
[001]
[100]
[010]
orientation
T
P
fO2
Fo
segment b
segment a
step
distance from rim
starting profile, measured Fo
b Forward model in time
starting profile
time
t1
t2
t3
Fo
√(D t)
distance from rim
c Misfit against time
1σ range
misfit χ2
t1
best-fit time
t3
time
noise model
correlated
white
distance from rim
d Uncertain starting profile
(optional; fitted with the time)
Fo
segment Fo
step position
distance from rim
e Monte Carlo
T
P
fO2
law parameters
log D
faster D,
shorter t
log t
t = (D t)fit / D
f Time distribution
median
fit alone
+ starting profile
+ T, P, fO2, law
probability
68% range
time (log scale)